

TIN200 2026

From Fragmented Evidence to a Governed Annual Ranking

Consultancy report for Technology Investment Network

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Executive Summary

Technology Investment Network (TIN) produces the annual TIN Report, ranking New Zealand's top 200 technology exporters by revenue and profiling the wider innovation ecosystem. The 2025 report recorded NZ\$20.0 billion in TIN200 revenue, NZ\$15.31 billion in exports, and 61,369 employees, making the integrity of the company's records material to investors, policymakers, and the Technology Investment Network (2025). This engagement supported the 2026 cycle by reviewing candidates, reconciling source lists, screening public evidence, checking ownership events and contacts, and consolidating reusable client records.

The analysis applied TIN's five published inclusion criteria, two operational screening controls, a four-tier evidence hierarchy, and structured reconciliation. It used the data-quality dimensions of accuracy, completeness, consistency, timeliness, and relevance (Wang & Strong, 1996), along with governance principles that assign ownership, establish standards, and define decision rights (Khatri & Brown, 2010). Because confidential survey responses and TIN's revenue ranking remain decisive, the outputs support decisions rather than confirm the 2026 list.

Six findings are supported by retained evidence. The directory contains 50 foreign-owned firms among 200 companies (25%), requiring consistent annual application of TIN's existing foreign-acquisition rule. Reconciliation retained 41 unmatched names across 93 source rows for review. Of 34 sponsor-provided companies, 16 were already in the TIN200, one appeared only in the emerging list, and 17 were not found. Four bounced-contact cases produced 14 alternatives, but no campaign denominator or observation period was retained, so neither a bounce rate nor a decay rate can be claimed. Although 30 acquisitions were reported as reviewed, only selected examples were supplied, requiring announcement, completion, and status to be recorded separately. Finally, 40 candidates were classified without an independent dual-review sample, so reviewer agreement was not measured.

The report recommends a lightweight, governed screening system. By 31 July 2026, TIN should implement a central candidate register containing stable identifiers, classifications, evidence links, confidence, reviewers, and review dates, supported by a shared dictionary, explicit exclusion reasons, pre-research matching, and dated contact checks. At least 10% of candidates should be dual-reviewed with a target agreement of 90% or more, and the foreign-acquisition decision tree should be operational by 31 August 2026. The one-click batch should generate standardized reporting inputs, after which staff manually refresh, validate, and publish the Power BI report. An annual economic scorecard should reproduce Appendix G's revenue growth, export intensity, scale, sector, and regional indicators from the cited official inputs.

Implementation begins with stable evidence and decision controls, preserving TIN's editorial judgement while improving auditability and reducing rework. Supplementary screening, contact, media monitoring, and cleantech outputs remain in Appendix H as separate pipeline evidence; they are not pooled with the audited 40-record population because the workstreams serve different purposes.

Contents

Executive Summary	2
Contents	3
1. Introduction.....	4
2. Project Background.....	5
2.1 TIN's annual reporting context.....	5
2.2 Engagement activities and deliverables	5
2.3 Managerial significance	6
3. Methodology and Analysis	6
3.1 Analytical framework	6
3.2 Unit of analysis and evidence base	7
3.3 Four-tier evidence hierarchy	7
3.4 Eligibility criteria and screening controls	8
3.5 Reconciliation and classification procedure	8
3.6 Contact and acquisition validation.....	9
3.7 Workflow and validation design.....	9
3.8 Limitations	10
4. Findings and Discussion	10
4.1 The central risk is fragmented decision evidence	11
4.2 Outreach suitability and final eligibility are different decisions	11
4.3 Reconciliation exceptions are measurable and should remain visible	11
4.4 List comparison reveals both overlap and a sizeable review queue.....	12
4.5 Contact evidence shows a timeliness risk, not a measured decay rate.....	12
4.6 TIN already has a foreign-acquisition rule; application is the gap	13
4.7 Reviewer consistency was not measured	14
4.8 Evidence retention is a management control	14
4.9 The findings from one control system	14
5. Conclusions.....	15
6. Recommendations and Implementation Roadmap.....	16
6.1 Create a central candidate register by 31 July 2026.....	16
6.2 Approve a data dictionary and status model by 15 July 2026.....	16
6.3 Record an explicit reason for every screening outcome by 31 July 2026.....	16
6.4 Introduce evidence and confidence rules by 31 July 2026	16
6.5 Revalidate selected survey contacts within 90 days of outreach	16

6.6 Match candidates before assigning research	17
6.7 Dual-review at least 10% of candidates from August 2026.....	17
6.8 Operationalize TIN's existing foreign-acquisition eligibility rule	17
6.9 Standardize one-click reporting preparation and manual Power BI publication	17
6.10 Establish an annual TIN200 economic context scorecard	17
6.11 Sequencing, assurance, and implementation risk.....	19
References.....	20
Appendices.....	21
Appendix A. Ordered engagement chronology	21
Appendix B. Screening record specification.....	22
Appendix C. Contact status definitions.....	22
Appendix D. Selected acquisition examples.....	23
Appendix E. Supporting automation and Power BI specification	24
Appendix E.1 One-click control flow	24
Appendix F. Dashboard views	25
Appendix G. Economic Context: The TIN200 and the New Zealand Economy	33
Appendix G.1 Sector Scale and Growth	33
Appendix G.2 Revenue Concentration and Scale.....	33
Appendix G.3 Sector and Regional Patterns.....	33
Appendix G.4 Derived Indicators and Interpretation.....	33
Appendix H. Supplementary Pipeline and Market Research Evidence.....	35
Appendix H.1 Supplementary workstream sequence	35
Appendix H.2 Population map and evidence boundaries	36
Appendix H.3 Broader candidate-screening profile.....	37
Appendix H.4 Contact-readiness profile.....	38
Appendix H.5 Media-monitoring outputs.....	38
Appendix H.6 Cleantech discovery universe.....	39
Appendix H.7 Recommended use and limitations.....	40

1. Introduction

Technology Investment Network is a specialist New Zealand research organization that tracks the country's technology exporters and publishes the annual TIN Report. Its central product is an evidence-based view of the sector: which firms qualify, how much revenue they generate, where they export, whom they employ, and how the ranking changes over time. TIN states that the report ranks the top 200 technology-exporting companies by revenue and identifies 100 early-stage firms encountered during the year (Technology Investment Network, n.d.). The value of the publication, therefore, depends on both confidential company returns and a reliable company master that can connect names, ownership, sectors, contacts, and evidence across reporting cycles.

The business problem is not a shortage of information. It is the difficulty of turning fragmented, uneven, and time-sensitive information into a defensible annual population. Private companies may publish little financial information; companies appear under trading names, legal names, and former names; acquisitions can change ownership without ending a company's New Zealand operations; and executive contacts can become obsolete between survey cycles. These conditions create several risks at once: an eligible company may be missed, the same company may be researched twice, an ownership event may be misinterpreted, or a survey may be sent to an outdated contact. Because TIN's data is part of the published product, there are editorial, reputational, and operational inefficiency risks.

The engagement objective was to strengthen the evidence and workflow used before the 2026 survey and ranking. The team reviewed candidate lists, classified potential entrants, compared sponsor and internal sources, researched acquisitions and company activity, recovered contacts, assembled contextual datasets, and consolidated the results into reusable client records. The intended outcome was a cleaner and more reviewable basis for TIN's decisions, while preserving TIN's editorial judgement.

The scope has three important boundaries. First, public evidence cannot reliably establish final revenue for many private firms, so this report does not claim to identify the final TIN200. Second, the analysis audits the retained project files; if the team did not preserve a denominator, a timestamped log, or a dual-review sample, the corresponding rate is reported as unavailable. Third, the team provided evidence and decision support rather than final eligibility rulings, which remain dependent on confidential company responses and TIN's internal review.

Two themes guide the report. The first is data quality as fitness for use: a record must be sufficiently accurate, complete, consistent, timely, and relevant for the decision being made. The second is data governance: the assignment of standards, decision rights, ownership, and review controls around shared data. Section 2 places the project in TIN's annual reporting process. Section 3 explains the evidence hierarchy, screening rules, and validation method. Section 4 presents six measured findings and their implications. Sections 5 and 6 draw the conclusions and set out an owned, timed, and measurable implementation roadmap.

2. Project Background

2.1 TIN's annual reporting context

TIN's annual publication is both a ranking and a sector dataset. Its 2025 release reported NZ\$20.0 billion in revenue, 9.9% annual growth, NZ\$15.31 billion in exports, and 61,369 employees. High-tech manufacturing contributed NZ\$9.80 billion, ICT NZ\$8.93 billion, and biotech NZ\$1.25 billion; Auckland accounted for NZ\$10.5 billion and Wellington NZ\$5.2 billion (Technology Investment Network, 2025). These figures show why company-level classification matters. A duplicate, omission, or ownership error can affect sector, regional, and ownership summaries as well as the published rank.

The formal entry rule is clearer than the earlier draft suggested. TIN publishes five inclusion criteria: a company must originate in New Zealand, retain a meaningful New Zealand presence, operate in high-tech manufacturing, ICT, or biotech, have developed its own technology-based intellectual property, and generate at least 10% of revenue offshore or plan to do so if it is an early-stage company. TIN also publishes a specific rule for foreign-acquired companies: they may remain included when the criteria are met, they have not been subsumed into the parent, and they still provide separate financial data (Technology Investment Network, n.d.). Accordingly, the project issue is consistent application and evidence retention, not an unresolved question about whether foreign ownership is permitted.

2.2 Engagement activities and deliverables

The eight-week engagement moved from population building to targeted validation and delivery. In the first weeks, the team reviewed the previous list, screened potential entrants, and compared sponsor suggestions with TIN and emerging-company lists. The middle weeks expanded the sector, investment, and export context, and supported outreach through contact research. Later work examined acquisitions, ownership, and executive contacts, then consolidated the review evidence and client-ready outputs. The detailed week-by-week chronology is moved to Appendix A so the main report can focus on analysis.

Table 1. Engagement workstreams and decision purpose

Workstream	Retained output	Decision supported
Candidate screening	40-company validation workbook	Keep, review, or remove a candidate before outreach
List reconciliation	34 sponsor entries compared with TIN200 and emerging lists	Prevent duplicate research and identify net-new candidates
Ownership and acquisitions	Review notes and selected acquisition evidence	Apply TIN's foreign-acquisition and continuity rules
Contact recovery	Four bounced-company cases and 14 alternatives	Improve survey reachability while preserving evidence quality
Evidence consolidation	Structured reconciliation records and validation summaries	Support repeatable exception review and client handover
Supplementary pipeline research	Aggregate screening, contact, media-monitoring, and cleantech summaries	Broaden future-candidate monitoring without treating pipeline signals as final eligibility

The outputs vary in evidential strength. Some are complete workbooks with countable records; others are selected examples or narrative logs. This distinction matters. A consultancy report should not convert activity into a measured result unless the underlying file supports the numerator, denominator, and status. The revised report, therefore, treats the directory, comparison workbook, candidate-validation workbook, bounce-recovery workbook, and reconciliation validation as quantified evidence. Acquisition examples are retained as illustrative because the complete 30-case status schedule was not available in the supplied project files.

2.3 Managerial significance

The project is best understood as a governance intervention around a recurring editorial process. A larger database alone would not solve the problem: more records without stable identifiers, ownership rules, and review dates would increase the work of reconciliation. Equally, a rules-only process would be unsafe because eligibility requires judgement about origin, presence, technology-based intellectual property, export intent, and the degree to which an acquired firm remains separately reportable. The practical opportunity is to standardize the evidence supporting judgement, make exceptions visible, and preserve the rationale for each decision.

This approach aligns with data-governance research. Khatri and Brown (2010) argue that governance concerns decision domains such as data principles, quality, metadata, access, and lifecycle, together with clarity about who is accountable. ISO 8000-61 similarly treats data-quality management as a process that can be specified, assessed, and improved (International Organization for Standardization, 2016). For TIN, this means the annual cycle should define who can create a candidate, who can resolve a name match, what evidence is sufficient, who can make an exception decision, and when a record expires or must be rechecked.

The economic significance of the work is summarized in Appendix G using verified TIN revenue, export, sector, regional, and employment figures. These measures show why consistent company classification matters to the reliability of the sector-level analysis produced from the annual ranking.

3. Methodology and Analysis

3.1 Analytical framework

The analysis used Wang and Strong's (1996) principle that data quality should be judged by data consumers, not only by technical correctness. Five dimensions were applied to the project. Accuracy asked whether a name, owner, contact, or status was supported by evidence. Completeness asked whether the fields needed for a decision were populated. Consistency was assessed by whether the same company and categories were consistently represented across sources. Timeliness asked when the evidence was last checked. Relevance asked whether a fact actually supported survey outreach, inclusion screening, or ranking. This framework prevents a common error: treating a syntactically clean spreadsheet as decision-ready when its evidence is incomplete or stale.

A governance lens was then used to translate the dimensions into controls. The proposed system requires a stable company identifier, a named data owner, controlled classifications, recorded evidence, a confidence rating, a reviewer, and a review date. It also preserves exceptions rather than deleting them. This is consistent with DAMA-DMBOK's emphasis on metadata, master and reference data, data quality, and stewardship as connected capabilities (DAMA International, 2017).

3.2 Unit of analysis and evidence base

The unit of analysis was the company or candidate-company record. Depending on the workstream, the assessed population was the 200-company directory, 34 sponsor-provided entries, 40 candidate-validation records, all source tables included in the reconciliation review, four known bounced-company cases, or the 30 acquisitions reported by the team as reviewed. These populations are not interchangeable. The revised findings always state the applicable denominator and do not generalize a four-case contact exercise to the whole TIN200.

The retained evidence included TIN_Companies.xlsx, the reconciliation validation output, TIN_Comparison_Output.xlsx, the other dataset, TIN_week1_40_validation_criteria_updated.xlsx, 2026_Bounced_Contact_Recovery_Tracking.xlsx, selected acquisition records, and the supplied consultancy-report draft. Official TIN pages were used to verify compliance with inclusion criteria and to obtain 2025 sector statistics. Company or regulatory evidence was preferred for ownership transactions. Where only selected examples were retained, the report labels them accordingly.

The broader team draft recorded additional populations: 127 working screening records, a 207-record comparison population of existing-list records, 63 refined media-monitoring records, and 92 cleantech records. These aggregates add useful context but were not independently reconciled with every row of the core audited evidence. Appendix H, therefore, reports them as supplementary workstream outputs and explicitly states their evidence status.

3.3 Four-tier evidence hierarchy

The earlier draft referred to a four-tier hierarchy but displayed five source categories. The corrected hierarchy contains four tiers, ordered by their ability to support a consequential classification. A lower tier can discover a lead or corroborate a claim, but it should not be the sole basis for a decisive ownership or eligibility conclusion.

Table 2. Four-tier source hierarchy

Tier	Source category	Examples	Permitted use
1	Regulatory and statutory	NZX, Companies Office, Overseas Investment Office, Takeovers Panel	Decisive evidence for legal entity, filing, transaction, and ownership status
2	Company official	Annual reports, company releases, investor pages, official websites	Primary evidence for products, leadership, operations, and announced transactions
3	Credible independent reporting	Radio New Zealand, New Zealand Herald, Reuters, and established business media	Corroboration, context, and discovery where primary records are limited
4	Discovery and triangulation	LinkedIn, PitchBook, Dealroom, Crunchbase, and search results	Lead generation and cross-checking only; not sole evidence for final inclusion or acquisition status

Sources were evaluated on relevance, authority, recency, and specificity. A company homepage that merely exists is weaker than a dated investor announcement addressing the exact transaction. Similarly, a media report is not a primary source, even when it is reputable. Acquisition records should therefore separate the announcement date, legal completion date, and current transaction status, and identify the evidence type rather than labeling every link 'primary'.

3.4 Eligibility criteria and screening controls

Candidate records were assessed against TIN's five published inclusion criteria. Two additional checks used during project work are screening controls, not extra TIN inclusion criteria. The first is the current operating status, used to avoid spending time on inactive or dissolved entities. The second is duplicate/list status, which indicates whether a company is already on the TIN200 or the emerging list. Revenue scale, funding, and market visibility may help prioritize outreach, but cannot substitute for the published criteria or TIN's confidential revenue evidence.

Table 3. Published inclusion criteria and operational screening controls

Type	Criterion or control	Evidence sought	Output
TIN criterion 1	Originated in New Zealand	Company history, statutory record, or official profile	Meets/does not meet/uncertain
TIN criterion 2	Meaningful New Zealand presence	Current staff, office, R&D, or operational evidence	Meets/does not meet/uncertain
TIN criterion 3	High-tech manufacturing, ICT, or biotech	Products, services, and sector evidence	Primary and secondary sectors
TIN criterion 4	Own technology-based intellectual property	Product, patent, platform, or R&D evidence	Meets/does not meet/uncertain
TIN criterion 5	At least 10% offshore revenue, or an early-stage plan to export	Survey response or credible export-market evidence	Meets/does not meet/requires TIN confirmation
Screening control 1	Current operating status	Companies Office, current website, and recent activity	Active/inactive/uncertain
Screening control 2	Duplicate and list status	Stable identifier, aliases, and existing lists	TIN200/emerging/new candidate/exception

3.5 Reconciliation and classification procedure

Name reconciliation followed a staged process. First, names were normalized for case, punctuation, spacing, and common legal suffixes. Second, exact and alias matches were checked against Company_Directory. Third, unmatched records were retained in Company_Master and Company_Name_Map for review rather than being forced into an uncertain match. This resembles record-linkage practice: a match is a probabilistic or evidence-based decision, not merely a string operation (Fellegi & Sunter, 1969). The practical output is reconciliation-ready data with visible exceptions.

Candidate classification used three operational outcomes: Keep, Review, or Remove. The 40-record validation workbook contained 31 Keep, six Review, and three Remove decisions. The corresponding TIN criteria screen contained 31 Likely Meets, six Needs Review, and three Likely Does Not Meet results. Confidence was recorded as High for 34 records after normalizing one capitalization typo and Medium for six. These counts describe the completed workbook; they do not demonstrate inter-rater reliability because no independent second rating was retained.

3.6 Contact and acquisition validation

Contact validation prioritized role-relevant people such as chief executives, finance leaders, and founders, then recorded alternative contacts where an existing address bounced. Four bounced-company cases were retained, with 14 alternatives compiled. A robust future log should also retain the send date, total contacts tested, successful deliveries, and bounce reason. Without those fields, the project can demonstrate contact risk but cannot calculate a bounce rate or the decay rate.

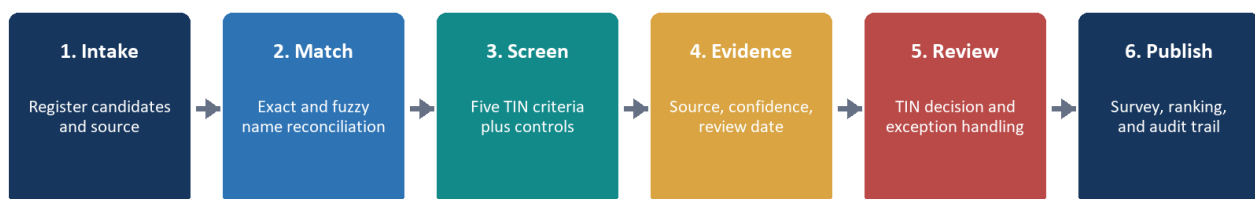
The acquisition validation asked three separate questions: when was the transaction announced, when did it legally complete, and what is its current status? It then considered whether the acquired company continued to meet TIN's five criteria, remained operationally distinct, and could provide separate financial data. The project reported 30 acquisitions reviewed, but the complete schedule was not present in the supplied report evidence. Appendix D, therefore, presents selected examples and explicitly directs the client to the confidential workbook for the full review.

3.7 Workflow and validation design

Figure 1. Proposed governed annual screening workflow

Figure 1 shows the proposed control sequence. Each stage leaves a record that can be reviewed later. Intake

Proposed governed annual screening workflow



captures provenance; matching prevents repeated work; screening separates TIN criteria from operational controls; evidence fields make confidence visible; TIN review resolves exceptions; and publication uses only approved records. This design preserves human judgement but makes it inspectable.

3.8 Limitations

Four limitations constrain interpretation. Public evidence is uneven, especially for private firms. The project cannot reproduce confidential survey evidence or final revenue ranking decisions. Several activity claims in the original draft lacked complete retained schedules, particularly acquisition status, contact denominators, and dual-review results. Finally, the complete acquisition status schedule, campaign-level contact denominator, and dual-review record were not available in the retained evidence. These limitations are not reasons to discard the work; they identify which controls and records the next cycle must preserve.

4. Findings and Discussion

The evidence supports six findings. Table 4 provides the results, denominators, and sources before the discussion interprets each finding. This structure is deliberate: managerial implications follow the measurement, and unavailable measurements are not reconstructed from memory.

Table 4. Findings Evidence Summary

Finding	Sample/denominator	Result	Evidence source
Foreign ownership	200 companies	50 foreign-owned; 25.0%	TIN_Companies.xlsx - Company_Directory
Reconciliation exceptions	All source tables included in the reconciliation review	41 distinct names across 93 source rows did not match Company_Directory; retained for review	Reconciliation validation output and name-mapping records
Acquisition review	30 acquisitions reported as reviewed	Complete announcement/completion/status totals were not retained in the supplied copy; selected examples only	Team narrative and selected acquisition evidence
Contact recovery	Four known bounced-company cases; campaign denominator not retained	14 alternative contacts compiled; bounce rate and decay period unavailable	2026_Bounced_Contact_Recovery_Tracking.xlsx
List comparison	34 sponsor-provided entries compared with a 143-company reference list	16 already in TIN200; 1 in emerging list only; 17 not found and routed to review	TIN_Comparison_Output.xlsx
Reviewer consistency	40 candidate records; no independently dual-reviewed sample	31 Keep, 6 Review, 3 Remove; agreement percentage not measured	TIN_week1_40_validation_criteria_updated.xlsx

4.1 The central risk is fragmented decision evidence

TIN's problem is not simply missing cells. Different workbooks answer different questions, and the link between a result and its evidence is not always preserved. A candidate can appear in a sponsor list under one name, in a directory under another, and in an acquisition note under a former owner. Unless those records share a stable identifier and decision history, a reviewer must reconstruct context each year. This creates inconsistency and avoidable labor even when each workbook is individually accurate.

The data-quality framework explains why. Completeness is decision-specific: a company name and website may be enough for discovery, but not for inclusion. Consistency requires common sector and status values across tables. Timeliness requires review dates for ownership and contacts. Relevance requires separating outreach potential from final eligibility. A central candidate register is therefore justified not as a fashionable database project, but as the smallest control capable of preserving these linked dimensions.

4.2 Outreach suitability and final eligibility are different decisions

The 40-record candidate workbook shows a useful screening funnel: 31 candidates were marked Keep, six Review, and three Remove. These labels support prioritization, but they do not prove that 31 firms belong in the next TIN200. A Keep decision means that available evidence justified continued research or outreach. Final eligibility still depends on all five criteria, company responses, and revenue ranking. The distinction is especially important for early-stage companies that may meet innovation and export-intent tests but do not yet meet the revenue threshold for the top 200.

Combining the two decisions into a single status would create false precision. A company may be an excellent survey candidate yet ultimately fall outside the revenue-ranked list; another may have weak public visibility but strong confidential revenue. The recommended register, therefore, stores both a screening disposition and a TIN decision. This allows outreach teams to act without misrepresenting the candidate list as a forecast of final membership.

4.3 Reconciliation exceptions are measurable and should remain visible

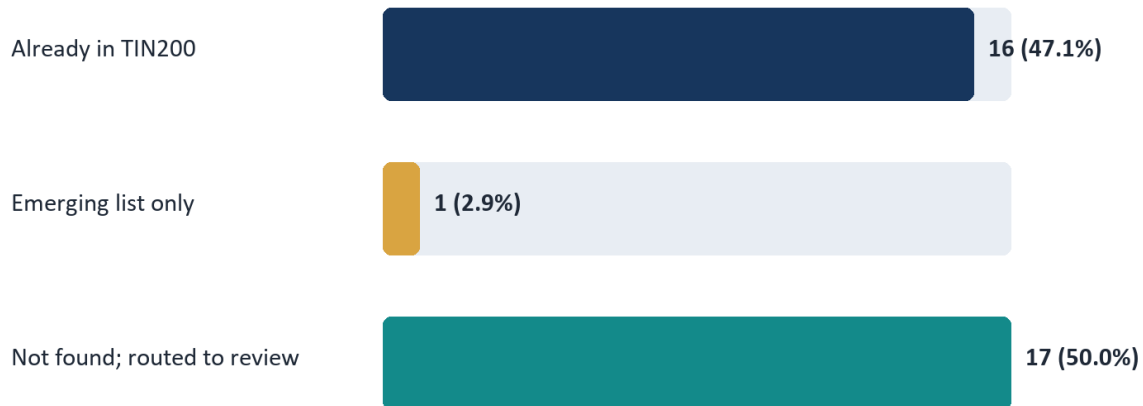
Reconciliation validation found 41 distinct source names across 93 source rows that did not match Company_Directory. These were not necessarily 41 new companies or 41 errors. They may include aliases, historical names, entities used only in a specialized source table, or genuine candidates missing from the directory. The important result is that the reconciliation process retained them as visible exceptions rather than dropping them or forcing a match. The correct description is therefore 'standardized and reconciliation-ready', not 'completely deduplicated'.

This exception set provides a concrete workload for stewardship. Each name must be assigned one of four resolutions: match an existing company identifier, create a new company record, classify as an intentional non-directory entity, or reject with a recorded reason. Once resolved, the alias or decision should persist so the same exception does not recur next year. A KPI based on unresolved exceptions is more meaningful than claiming that a data-cleaning step removed all duplication.

4.4 List comparison reveals both overlap and a sizeable review queue

Figure 2. Disposition of the 34 sponsor-provided companies

Disposition of 34 sponsor-provided companies



Source: TIN_Comparison_Output.xlsx, Sponsor_NonSponsor_List. Denominator: 34 sponsor entries.

The verified comparison of 34 sponsor entries against the 143-company reference list does not support the earlier statement that only three companies overlapped. The retained workbook shows 16 already in the TIN200, one in the emerging list only, and 17 not found in either list. The first group represents duplicate research that can be avoided through an intake match. The single emerging-list match requires a status update rather than a new discovery. The remaining 17 are a genuine review queue, not evidence that they qualify.

The managerial implication is two-sided. Almost half of the sponsor entries were already in the principal list, demonstrating the value of matching before assigning research. At the same time, half were not found, showing that sponsor and external channels contribute to useful discovery. The appropriate response is not to reject duplicate-rich sources, but to make provenance and match status visible at intake so that research effort is directed to net-new or materially changed companies.

4.5 Contact evidence shows a timeliness risk, not a measured decay rate

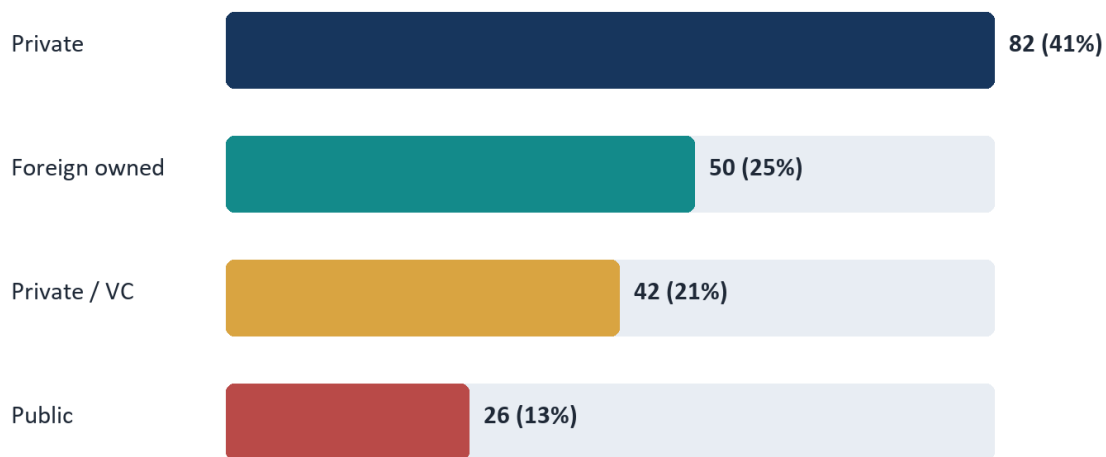
Four bounced-company cases were retained for Oritain, FrameCAD, Futura Trails/Futura Trailers, and Argenta. The recovery workbook contains 14 alternative contacts, averaging 3.5 per affected company. This is useful operational evidence that contact records can fail and that fallback contacts can be assembled. It does not establish how quickly contact data decays. The project did not retain the total number of emails sent, the period over which they were tested, or a complete delivery result, so no valid bounce percentage can be calculated.

The control should match the evidence. Contact records need a verified-on date, source, role, direct-versus-generic status and latest delivery outcome. Before each annual survey, contacts selected for outreach should be rechecked within a defined window. Measuring successful delivery and the bounce reason would then allow TIN to calculate its own decay rate over time, rather than relying on generic vendor claims.

4.6 TIN already has a foreign-acquisition rule; the application is the gap

Figure 3. Ownership composition of the 200-company directory

Ownership composition of the 2025 TIN200 directory



Source: Company_Directory, TIN_Companies.xlsx. Denominator: 200 companies.

The directory contains 50 foreign-owned companies out of 200, or 25%. This is material, but it does not create an unresolved policy choice. TIN's published rule already allows foreign-acquired firms when they continue to meet the five criteria, have not been subsumed into the parent, and provide separate financial data (Technology Investment Network, n.d.). Ownership type should therefore trigger a structured review, not automatic removal or a new debate about eligibility.

The practical challenge is evidence. 'Foreign-owned' alone does not reveal whether the New Zealand company retains meaningful operations, controls, or develops technology-based intellectual property, or reports financials separately. These are time-sensitive conditions. A company may qualify immediately after an acquisition and later become integrated into the parent. The recommended annual decision tree records the announcement and completion dates, current operating status, separate reporting evidence, criterion-by-criterion assessment, decision owner, and next review date.

Acquisition evidence also needs cleaner terminology. The supplied draft stated that all 30 acquisitions were documented, while the appendix displayed eight examples and mixed media with primary sources. The complete 30-case schedule was not present in the files audited for this report. Appendix D is therefore labeled 'selected acquisition examples'. It separates announcement, completion, and status, and it does not infer a completion date from an announcement.

4.7 Reviewer consistency was not measured

The 40-record validation workbook demonstrates that a shared classification structure was used, but it does not demonstrate consistency between reviewers. Thirty-one records were classified as Keep, six as Review, and three as Remove; confidence was High for 34 and Medium for six after normalizing a capitalization typo. However, the project retained no independent second ratings, disagreement log, or adjudication outcome. An agreement percentage cannot be calculated from a single final classification.

This distinction strengthens rather than weakens the recommendation. Manual judgement is necessary because the criteria require interpretation, but its reliability should be tested. A 10% dual-review sample in the next cycle would provide a manageable check. Agreement should be calculated before adjudication, with differences grouped by criterion. A target of at least 90% overall agreement is proposed; repeated disagreement on a particular criterion should trigger an update to the definition or evidence rule.

4.8 Evidence retention is a management control

The project shows that the strength of a finding depends on the evidence retained after the research activity is complete. The directory, comparison workbook, and candidate-validation workbook support quantified conclusions because their populations and classifications can be inspected. By contrast, the acquisition, contact, and reviewer-consistency claims are necessarily narrower where the complete schedule, campaign denominator, or independent ratings were not preserved. Evidence retention is therefore a management control, not an administrative afterthought.

TIN should define a compact evidence pack for each annual workstream. At minimum, it should preserve the assessed population, decision categories, supporting links, review dates, exceptions, and approval status. This would allow future reports to distinguish completed work from selected examples and to calculate rates only when a valid denominator exists. A monthly evidence-completeness check would also identify missing schedules early enough for the team to correct them before final reporting.

4.9 The findings from one control system

The six findings should not be managed as separate clean-up tasks. They describe failures at successive points in the same information lifecycle. List overlap begins at intake when provenance and matching are weak. Reconciliation exceptions persist when identity decisions are not stored. Screening uncertainty grows when the evidence required for each criterion is unclear. Contact failures appear when time-sensitive fields lack renewal dates. Acquisition ambiguity remains when announcement, completion, and current operating status are combined. Reviewer consistency cannot be assessed when independent decisions are overwritten by a single final answer. A control applied only at the end would therefore detect symptoms but not prevent their recurrence.

The most effective intervention point is intake. A stable identifier and source record allow TIN to recognize that a candidate has already been considered, preserve aliases, and distinguish a sponsor resubmission from a genuinely new company. The second intervention point is decision capture. Criterion-level statuses, evidence, and confidence allow us to determine whether uncertainty concerns New Zealand origin, operating presence, technology, intellectual property, export activity, or simply missing public information. The third is renewal: contacts, ownership, and acquisition conditions require a next-review date because correctness at one point in time does not guarantee correctness during the next survey cycle.

This integrated view also clarifies priority. The central register, data dictionary, and matching controls are foundational because every other measure depends on them. Contact revalidation and foreign-acquisition review are then recurring operational controls. Dual review is an assurance control that tests whether the method is being applied consistently. Management reporting is last in the dependency chain: it can communicate quality and exceptions effectively only after the underlying statuses and ownership responsibilities are defined. Presenting results before those controls are established would make incomplete governance more visible, but not more reliable.

The analysis, therefore, supports a proportionate response. TIN does not need to eliminate judgement, purchase an enterprise master data platform, or conduct exhaustive research on every lead. It needs enough structure to route low-confidence cases to review, enough history to avoid repeating work, and enough measurement to know whether quality is improving. This is why the recommendations use completeness, unresolved-exception, contact-currency, and reviewer-agreement indicators rather than broad promises of efficiency.

The supplementary screening and market-research workstreams reinforce this proportionate approach. Their value is in expanding TIN's discovery pipeline, identifying records requiring review, and preserving time-sensitive company developments. They do not alter the six principal findings or establish that a monitored company meets TIN's revenue-ranked inclusion threshold.

5. Conclusions

The engagement produced useful candidate, contact, ownership, reconciliation, and reporting assets, but its most important conclusion is procedural. TIN's annual screening process needs a single governed record of what was considered, what evidence was used, who decided, and when the decision must be revisited. The retained workbooks show why: 41 names remain unresolved against the directory, half of the 34 sponsor entries formed a new review queue, four contact failures required recovery, and 25% of the directory is foreign-owned and therefore subject to an existing conditional rule.

The evidence also defines the boundary of responsible reporting. The candidate shortlist is not a forecast of the next TIN200; selected acquisition examples are not a complete transaction register; four contact failures do not establish a decay rate; and a final classification does not measure reviewer agreement. By making these limits visible, TIN can focus the next cycle on retaining the denominators and reviewing the records needed for stronger decisions.

The recommended response is a lightweight governance process built around structured evidence and periodic quality review. It preserves TIN's editorial judgement while standardizing intake, matching, evidence, confidence, contact currency, and exception review. Reporting outputs should remain supporting tools: their purpose is to communicate approved results and visible exceptions, not to conceal uncertainty behind polished presentation.

Appendix G provides economic context for these recommendations through TIN's verified 2025 figures and transparent derived indicators. The scale of the population reinforces the importance of accurate underlying records to New Zealand's technology-sector intelligence.

6. Recommendations and Implementation Roadmap

The ten recommendations below are specific, measurable, achievable, relevant, and time-bound. Owners are role-based because TIN may assign named staff after accepting the report. Dates assume implementation begins after submission and before the next reporting cycle.

6.1 Create a central candidate register by 31 July 2026

The TIN Research Lead should implement a single controlled candidate database that covers discovery, matching, screening, outreach, and the final decision. Every candidate record should contain a stable identifier, a canonical name, aliases, a source channel, a classification, a confidence, an evidence link, a reviewer, a review date, and a next action. Completion means 100% of active candidates have all mandatory governance fields or a documented exception. This directly addresses fragmented decision evidence and recurring name reconciliation.

6.2 Approve a data dictionary and status model by 15 July 2026

The Head of Research should approve definitions for candidate, TIN200, emerging, Keep, Review, Remove, active, inactive, foreign-owned, acquired, announced, and completed. The dictionary should identify which values are TIN inclusion decisions and which are operational controls. All staff working on the 2026 cycle should complete a short calibration exercise, and 100% of controlled fields should use approved values rather than free-text variants.

6.3 Record an explicit reason for every screening outcome by 31 July 2026

The Data Steward should require each Keep, Review, and Remove result to include the relevant criterion, a short rationale, and evidence. The target is 100% reason coverage and zero blank decisions at the point of client review. This makes the funnel auditable and prevents a candidate from being excluded merely because it has weak public visibility.

6.4 Introduce evidence and confidence rules by 31 July 2026

TIN should adopt the four-tier hierarchy in Table 2. A High-confidence consequential decision should be supported by a Tier 1 or Tier 2 source, or by two independent sources where primary evidence is unavailable. The monthly quality check should target at least 95% compliance and list all exceptions for Research Lead approval. Media and commercial databases should be identified as secondary or discovery sources, not as primary sources.

6.5 Revalidate selected survey contacts within 90 days of outreach

The Outreach Lead should separate company eligibility from contact readiness and require each selected contact to have a role, source, verified-on date, and direct-versus-generic flag. Before the next survey send, 100% of selected contacts should have been checked within the previous 90 days. Delivery outcomes must be logged so that TIN can calculate a valid bounce rate and compare it across cycles.

6.6 Match candidates before assigning research

The Data Steward should run exact-name, alias, and probable-match checks at intake. The target is zero unresolved exact duplicates before deep research begins, while probable matches remain visible for human review. The 41 unmatched names already identified should be resolved or assigned a documented exception status by 31 August 2026. Alias decisions must persist into the next cycle.

6.7 Dual-review at least 10% of candidates from August 2026

The Research Lead should independently assign two reviewers to a stratified sample covering sectors, ownership types, and decision outcomes. Agreement should be measured before adjudication, with a target of at least 90%. Any criterion below 85% agreement should be discussed, and its guidance amended before final survey outreach. This creates the evidence of reviewer consistency that the current project lacks.

6.8 Operationalize TIN's existing foreign-acquisition eligibility rule

By 31 August 2026, the Research Lead should implement a documented decision tree and annual evidence review for every foreign-owned or newly acquired company. The review must test the five published criteria, whether the company has been subsumed into its parent, and whether separate financial data remains available. The target is 100% of foreign-owned directory records carrying a current decision, evidence link, reviewer, and next-review date.

6.9 Standardize one-click reporting preparation and manual Power BI publication

Before each annual dashboard release, the Reporting Lead should run the one-click batch file to execute the approved preparation scripts and generate the standardized Excel outputs used by the preconfigured Power BI model. A staff member should then open the Power BI Desktop file, refresh and validate the model, check key totals and visuals against the generated outputs, and manually publish the approved report to the Power BI Service. Completion should require a successful batch run, the generation of all required output tables, zero unresolved refresh errors, documented validation of control totals, and confirmation of manual publication. This preserves human quality assurance at the publication point while reducing repetitive preparation work.

6.10 Establish an annual TIN200 economic context scorecard

By 30 September 2026, the TIN Insights Lead should establish a controlled annual economic scorecard using TIN's official published figures and the calculation schedule in Appendix G. It should report total revenue and annual growth, a comparable long-run revenue growth rate, export intensity, revenue per employee as a scale indicator, the share of companies above NZ\$100 million, primary-sector shares, and selected regional shares. Every indicator should record its source year, inputs, formula, definition, and reviewer, and reconcile to the approved report totals. The target is 100% cited and reproducible indicators, with no unsupported claims of productivity, concentration, or forecasting. The scorecard should inform annual commentary and management review without replacing company-level revenue verification.

Table 5. SMART implementation roadmap

Priority	Action	Owner	Deadline	KPI	Target	Dependency
P1	Central candidate register	TIN Research Lead	31 Jul 2026	Mandatory-field completeness	100% active candidates complete or exceptions recorded	Approved schema and access
P1	Data dictionary and status model	Head of Research	15 Jul 2026	Controlled-field adoption	100% approved values; all reviewers calibrated	Client approval
P1	Screening rationale	Data Steward	31 Jul 2026	Decisions with reason and evidence	100%	Dictionary and evidence fields
P2	Evidence and confidence rules	Research Lead	31 Jul 2026	High-confidence decisions meeting rule	At least 95%	Four-tier hierarchy
P1	Contact verification	Outreach Lead	Before next survey send	Selected contacts checked in prior 90 days	100%; delivery outcomes logged	Outreach list and send log
P1	Pre-research matching	Data Steward	31 Aug 2026	Unresolved exact duplicates; old exceptions	0 exact duplicates; all 41 exceptions resolved or classified	Stable IDs and aliases
P2	Dual-review sample	Research Lead	From Aug 2026	Independent agreement	At least 10% sampled; at least 90% agreement	Two trained reviewers
P1	Foreign-acquisition decision tree	Research Lead	31 Aug 2026	Foreign-owned records with current review	100%	Published TIN rule and acquisition evidence
P2	One-click reporting preparation and manual Power BI publication	Reporting Lead	Before each annual dashboard release	Successful batch run, Power BI validation and publication checklist	100% required outputs generated; report manually verified and published	Stable schemas, file locations and configured PBIX source paths
P2	Annual TIN200 economic context scorecard	TIN Insights Lead	30 Sep 2026 and before each report release	Indicators with cited inputs, formulas and reviewer	100% reproducible and reconciled to approved totals	Official TIN figures and Appendix G calculation schedule

Implementation should begin with the dictionary and register because the remaining recommendations depend on common fields. TIN can initially use a controlled spreadsheet or SharePoint list rather than purchasing a new platform. The design should be judged by whether it preserves identifiers, validation rules, evidence links, change history, and role-based ownership. Once the governed inputs are stable, the one-click batch should generate the standardized reporting files for the configured Power BI model. Publication should remain a controlled manual step: staff open the PBIX file, refresh it, validate control totals and visuals, and publish the approved report through Power BI Desktop. The same approved totals should populate the annual economic context scorecard so that dashboard reporting and published economic commentary remain aligned.

6.11 Sequencing, assurance, and implementation risk

The roadmap should be delivered in three short phases. Phase 1, during July 2026, establishes the dictionary, register, mandatory fields, and evidence rules. A small pilot using the 40 already classified candidates should be migrated first, allowing TIN to test validation lists, aliases, and review fields before loading the wider population. Phase 2, by the end of August, resolves the 41 reconciliation exceptions, applies the acquisition decision tree to the 50 foreign-owned directory records, and introduces the dual-review sample. Phase 3 embeds the controls in outreach and reporting, records delivery outcomes, runs the one-click preparation workflow, manually validates and publishes the refreshed Power BI report, updates the annual economic context scorecard, and presents the quality indicators in the monthly management review.

Three implementation risks require active management. The first is over-design: too many mandatory fields would encourage blank or low-quality entries. TIN should begin with the fields in Appendix B and add fields only when they support a recurring decision. The second is ownership ambiguity. A shared file without a named Research Lead and Data Steward would reproduce the current fragmentation, so roles must be assigned before migration. The third is premature rules-based acceptance. Probable name matches, foreign-acquisition eligibility, and low-confidence criterion assessments should remain review queues; automatic acceptance would create hidden errors.

Assurance should be lightweight but routine. The Research Lead should review the monthly quality scorecard, sample evidence links, and sign off on unresolved exceptions before survey release. At the end of the reporting cycle, TIN should compare the candidate register with the final survey and ranking outcomes: how many Keep records responded, how many reached the ranked population, which exclusions changed, and where evidence was insufficient. That feedback converts the register from an archive into a learning system and provides a defensible basis for adjusting criteria guidance in the following year.

Progress should be reviewed monthly through a short quality scorecard showing active candidates, mandatory-field completeness, unresolved matches, contacts due for recheck, acquisition reviews due, dual-review agreement, and the latest reconciliation result. These indicators create the evidence needed to evaluate whether the recommendations reduce rework and improve auditability in the next reporting cycle.

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Appendices

The appendices contain operational detail that supports, but is not required to follow, the managerial argument.

Appendix A. Ordered engagement chronology

Table A1. Eight-week project chronology

Week	Principal activity	Retained or reported output
1	Reviewed the prior TIN200 and screened potential entrants	Candidate-validation workbook and initial removal/review logic
2	Compared sponsor recommendations and existing lists	Sponsor/non-sponsor comparison and net-new review queue
3	Expanded candidate and sector research	Additional company evidence and shortlist development
4	Developed cleantech, investment, export, and contact-recovery inputs	Context datasets and four bounced-company cases
5	Expanded new-entrant contacts and outreach materials	Contact research and executive communication support
6	Updated role-relevant contacts for TIN companies	Contact records for client review; complete send denominator not retained
7	Reviewed acquisitions and ownership changes	Thirty cases reported as reviewed; selected examples retained in the supplied report evidence
8	Consolidated outputs, automation, and dashboard	Standardized Excel outputs, validation file, and Power BI report

Appendix B. Screening record specification

Table B1. Minimum fields for the central candidate register

Field group	Required fields	Purpose
Identity	Company_ID; canonical name; legal name; aliases; website	Stable matching across cycles and workbooks
Provenance	Source channel; date added; submitted by; original list	Preserve where the candidate came from
TIN criteria	Five criterion statuses and criterion-level evidence	Separate published eligibility tests from general attractiveness
Controls	Operating status; TIN200/emerging/duplicate status	Prevent wasted research and flag inactive entities
Decision	Keep/Review/Remove; TIN final decision; rationale	Separate research disposition from final eligibility
Evidence	URL or document reference; source tier; evidence date; confidence	Make claims reviewable and time-bound
Governance	Reviewer; review date; approver; next-review date	Assign accountability and renewal
Outreach	Contact; role; email; direct/generic; verified date; delivery outcome	Measure contact currency and campaign performance

Appendix C. Contact status definitions

Table C1. Recommended contact-quality statuses

Status	Definition	Required next action
Verified direct	Named role-relevant contact confirmed from a current source within 90 days	Use for outreach and record delivery results
Verified generic	Current company inbox or form confirmed, but no named decision-maker	Use as a fallback and continue direct-contact research
Unverified	Contact found, but recency, role, or source is insufficient	Do not use until checked
Bounced	Delivery failure recorded	Record the bounce reason and begin recovery
Departed / role changed	Evidence indicates the person is no longer suitable	Retain historical record; identify replacement
No contact found	Search completed without a supportable contact	Escalate to phone or the client relationship owner

Appendix D. Selected acquisition examples

The project narrative states that 30 acquisitions were reviewed. The complete, confidential 30-case evidence workbook was not included in the supplied report files; this appendix provides only selected examples. Before the client uses it, the complete workbook should contain separate announcement, completion, and current-status fields for every case. Media sources should be labeled secondary rather than primary.

Table D1. Selected acquisition examples and evidence status

Company/acquirer	Announcement date	Completion date	Status in retained evidence	Evidence
Robotics Plus / Yamaha Motor	25 Feb 2025	Not stated on the cited page	Agreement announced; Tauranga operations, jobs, and IP described as continuing	Robotics Plus official announcement: https://www.roboticsplus.co.nz/news/robotics-plus-acquisition-by-yamaha-motor-to-enable-precision-agriculture-for-growers/
Smartpay / Shift4	Reported in team log	Not independently confirmed in the supplied evidence	Requires confirmation against the relevant NZX scheme and implementation filings	Not available in the supplied evidence; consult the confidential acquisition workbook
Volpara / Lunit	Reported in team log	Not independently confirmed in the supplied evidence	Requires confirmation against company, market, or takeover records	Not available in the supplied evidence; consult the confidential acquisition workbook
Endace / management ownership	Historical ownership correction reported	Not applicable to current-year transaction	Treat as ownership-history correction; current separate-reporting evidence still required	Not available in the supplied evidence; consult the confidential acquisition workbook

Appendix E. Supporting automation and Power BI specification

The automation supports repeatable preparation and review. It should be described as standardization and reconciliation support, not complete deduplication or unattended Power BI Service refresh. Technical reference: Microsoft. (2026). Data refresh in Power BI. Microsoft Learn. <https://learn.microsoft.com/en-us/power-bi/connect-data/refresh-data>

Table E1. Current technical workflow and verified status

Component	Current function	Verified status/limitation
RUN_ONE_CLICK.bat	Runs tin_powerbi_prepare.py and then tin_revenue_estimation.py	Both Python stages are invoked by the batch file
tin_powerbi_prepare.py	Builds stable Excel tables and validation outputs for Power BI	Standardizes schemas and exposes reconciliation exceptions
tin_revenue_estimation.py	Produces supporting revenue-estimation output	Separate analytical stage; results require business review
Power BI model	Reads stable table and file names for visualization	Desktop refresh supported after source-path configuration
Power BI Service publication	Staff open the PBIX file after the one-click preparation run, refresh and validate the report, then publish it manually	Manual at present; publication follows desktop validation of generated outputs
Reconciliation	Compares source names with Company Directory and retains mappings/exceptions	41 names across 93 rows remain outside the directory for review
Runtime and reproducibility tests	Should record duration, row counts, exception counts, and file checks	No retained benchmark log or byte-identical hash test was available

Appendix E.1 One-click control flow

RUN_ONE_CLICK.bat

1. Execute tin_powerbi_prepare.py
2. Execute tin_revenue_estimation.py
3. Write standardized Excel and validation outputs
4. Refresh Power BI Desktop / Service manually

tin_pipeline_config.json

"trigger_powerbi_refresh": false

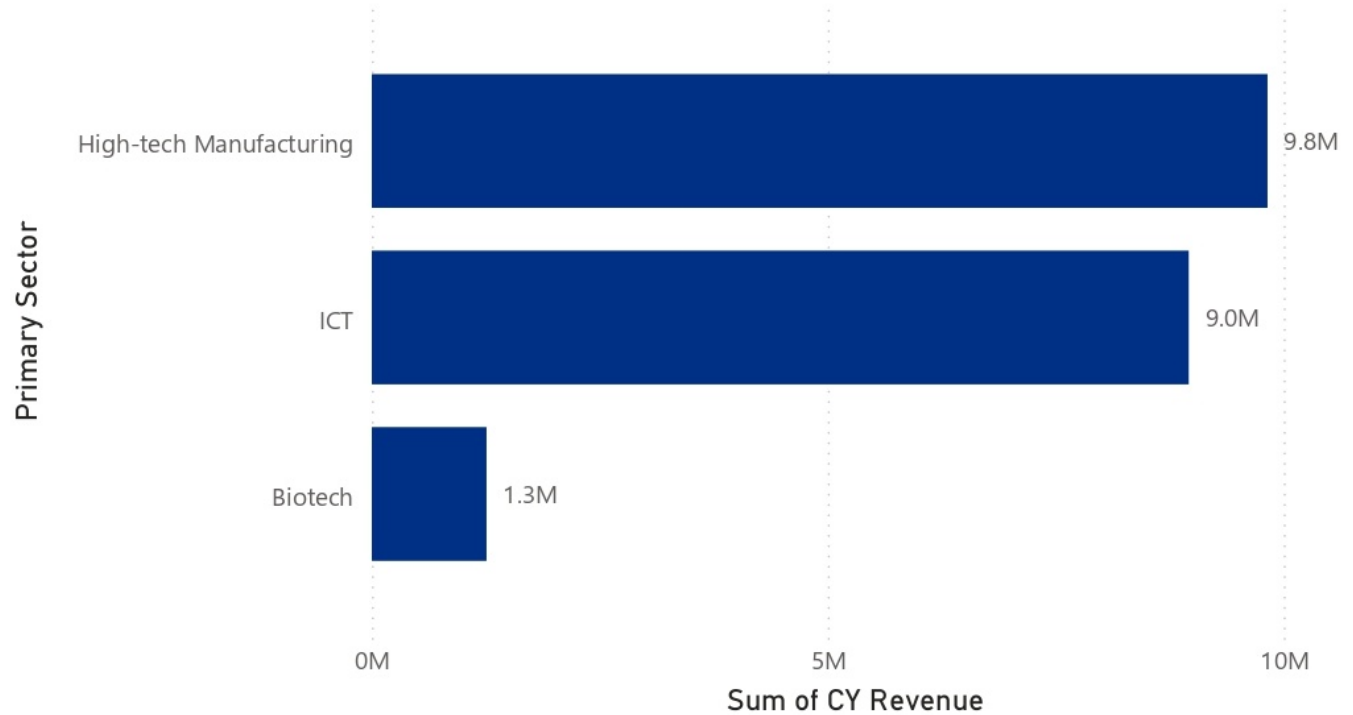
Appendix F. Dashboard views

Figures F1-F8 document the principal Power BI views developed during the engagement. They provide company-level auditability and comparison across ranking, sector revenue, ownership, market destination, and workforce dimensions. The figures are included as supporting technical evidence and do not replace the quantified findings in the main report.

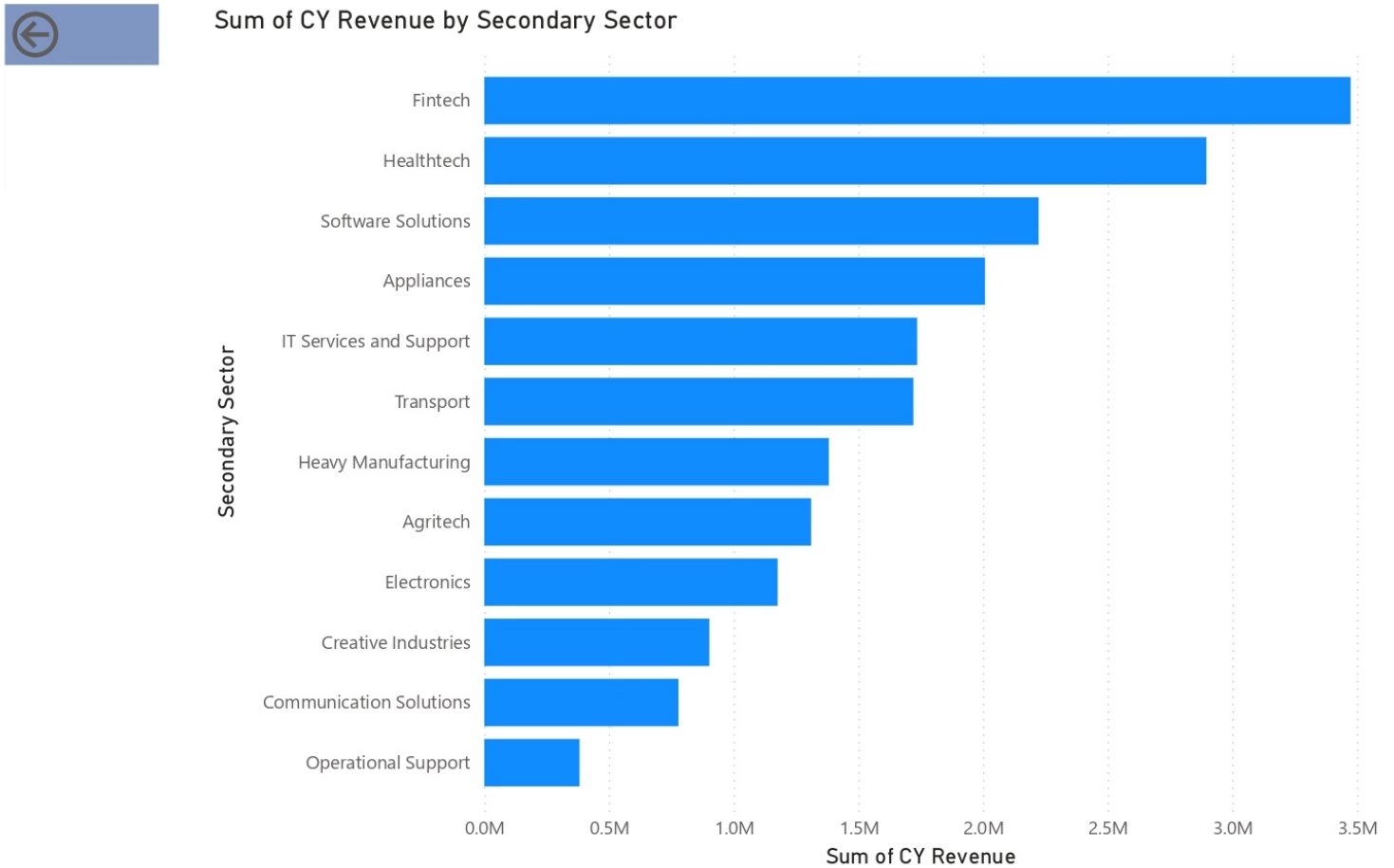
Figure F1. Company ranking and audit table

Sum of CY Ranking	Name	Primary Sector	Sum of CY Revenue (\$m)	Ownership	Region	Sum of Revenue_Growth_Pct
1	Xero	ICT	2,102.70	Public	Wellington	22.69
2	Fisher & Paykel Healthcare	High-tech Manufacturing	2,021.00	Public	South Auckland	15.96
3	Fisher & Paykel Appliances	High-tech Manufacturing	1,617.00	Foreign Owned	South Auckland	7.73
4	Datacom Group	ICT	1,483.00	Private	Wellington	0.89
5	Rocket Lab	High-tech Manufacturing	745.90	Public	South Auckland	78.34
6	Tait Communications	High-tech Manufacturing	516.00	Private/VC	Canterbury / Upper South Island	22.68
7	Weta FX	ICT	478.00	Private	Wellington	11.07
8	Gallagher Group	High-tech Manufacturing	441.40	Private	Hamilton	9.69
9	Windcave	ICT	404.00	Private	South Auckland	32.46
10	Pushpay	ICT	375.00	Foreign Owned	South Auckland	2.19
11	Livestock Improvement Corporation	Biotech	295.10	Public	Hamilton	10.41
12	Seequent	ICT	287.00	Foreign Owned	Canterbury / Upper South Island	7.38
13	Scott Technology	High-tech Manufacturing	275.00	Public	Otago / Southland	-0.41
14	Fusion5	ICT	245.00	Private/VC	Wellington	36.11
15	Gentrack Group	ICT	230.00	Public	South Auckland	7.88
16	Tira Group Limited	High-tech Manufacturing	210.00	Private	Hamilton	2.44
17	AFT Pharmaceuticals	Biotech	208.10	Public	North Auckland	6.49
18	Temperzone Group	High-tech Manufacturing	206.00	Private	South Auckland	4.65
19	Douglas Pharmaceuticals	Biotech	205.40	Private	South Auckland	-4.66
20	EROAD	ICT	194.40	Public	North Auckland	6.81
21	Buckley Systems	High-tech Manufacturing	190.00	Private	South Auckland	4.17
22	Daifuku Oceania	High-tech Manufacturing	185.10	Foreign Owned	North Auckland	-11.39
23	Moffat Group	High-tech Manufacturing	176.60	Foreign Owned	Canterbury / Upper South Island	23.79
24	Action Manufacturing	High-tech Manufacturing	165.70	Private	North Auckland	-7.17
25	Argenta	Biotech	162.90	Foreign Owned	South Auckland	0.00
26	MHM Automation	High-tech Manufacturing	162.00	Foreign Owned	Canterbury / Upper South Island	13.69
20087			19,990.30			2,456.46

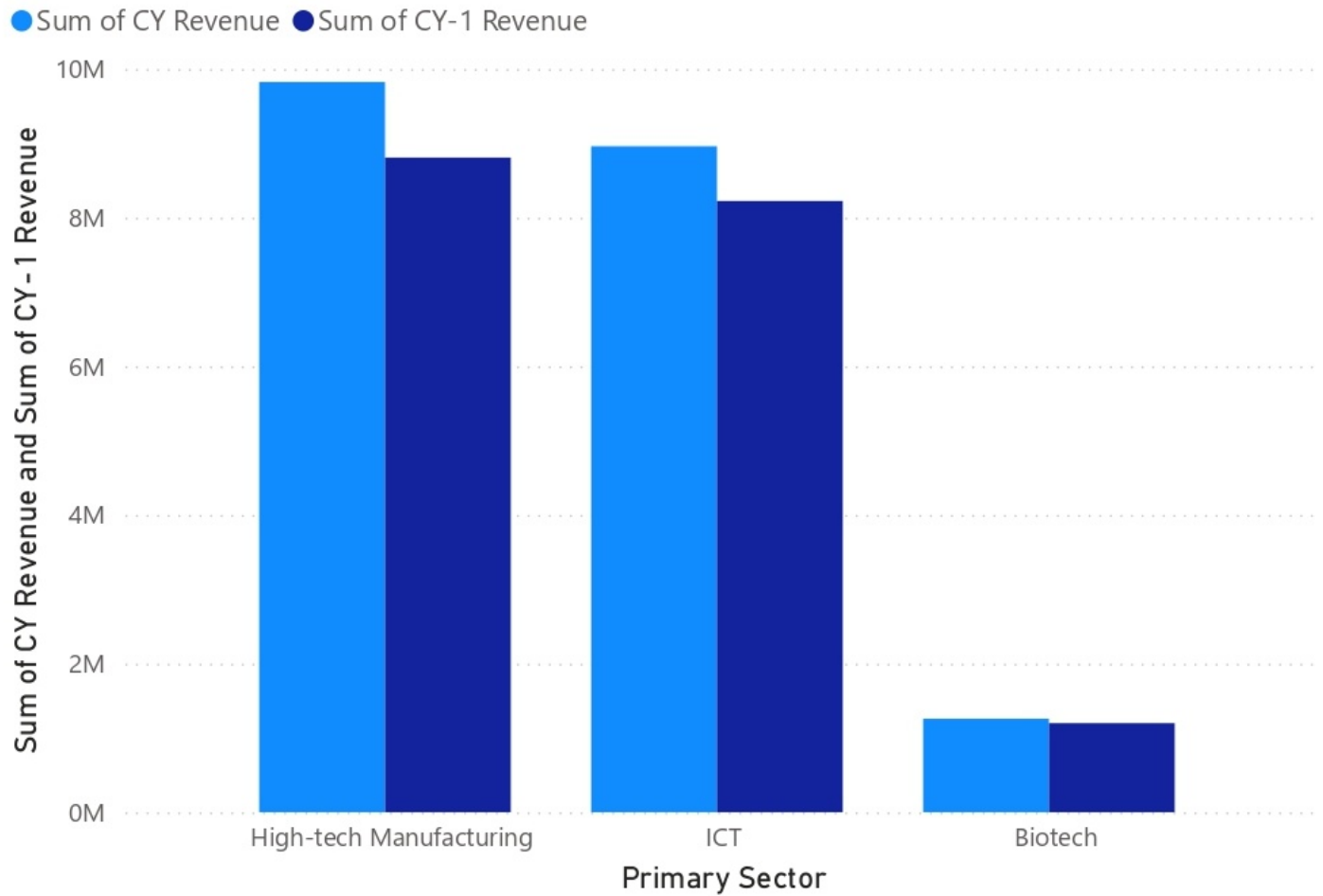
Interpretation. The company-level view brings the main ranking and classification fields together, allowing reviewers to trace rank, revenue, ownership, region, and year-on-year growth for individual companies.

Figure F2. Total 2025 revenue by primary sector**Total 2025 Revenue by Primary Sector (NZD \$m)**

Interpretation. High-tech manufacturing is the largest primary sector, followed by ICT and biotech. Power BI abbreviates the plotted totals using automatic display units; the bars correspond to approximately NZ\$9.8 billion, NZ\$9.0 billion, and NZ\$1.3 billion.

Figure F3. Total 2025 revenue by the secondary sector

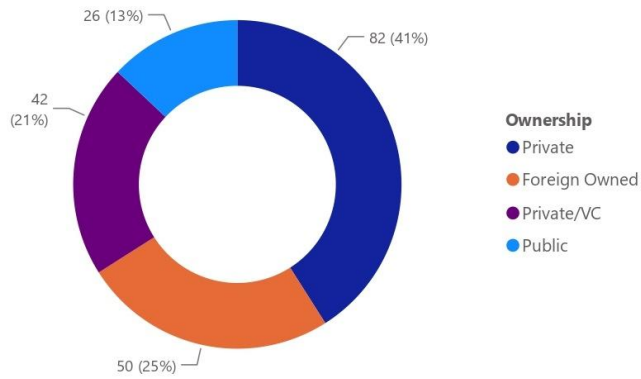
Interpretation. The secondary-sector view provides a more granular breakdown than the three primary sectors. Fintech and healthtech appear as the largest sub-sector groupings in the displayed data.

Figure F4. Primary-sector revenue comparison, 2024 and 2025**Revenue by Sector: 2024 vs 2025 (NZD \$000)**

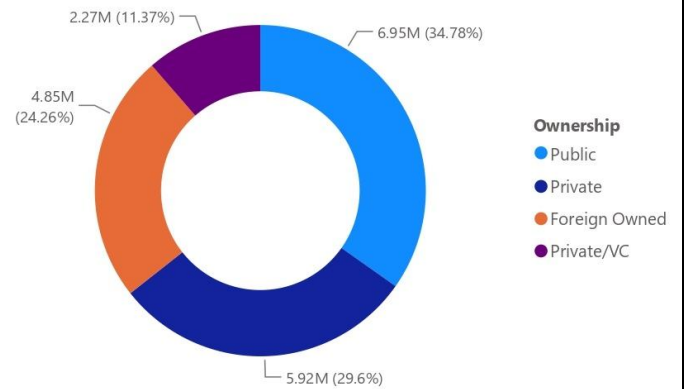
Interpretation. All three primary sectors are higher in 2025 than in the prior-year comparison shown. The chart uses NZD thousands, so an axis value of 10M represents approximately NZ\$10 billion.

Figure F5. Ownership composition by company count and revenue

Company Count by Ownership



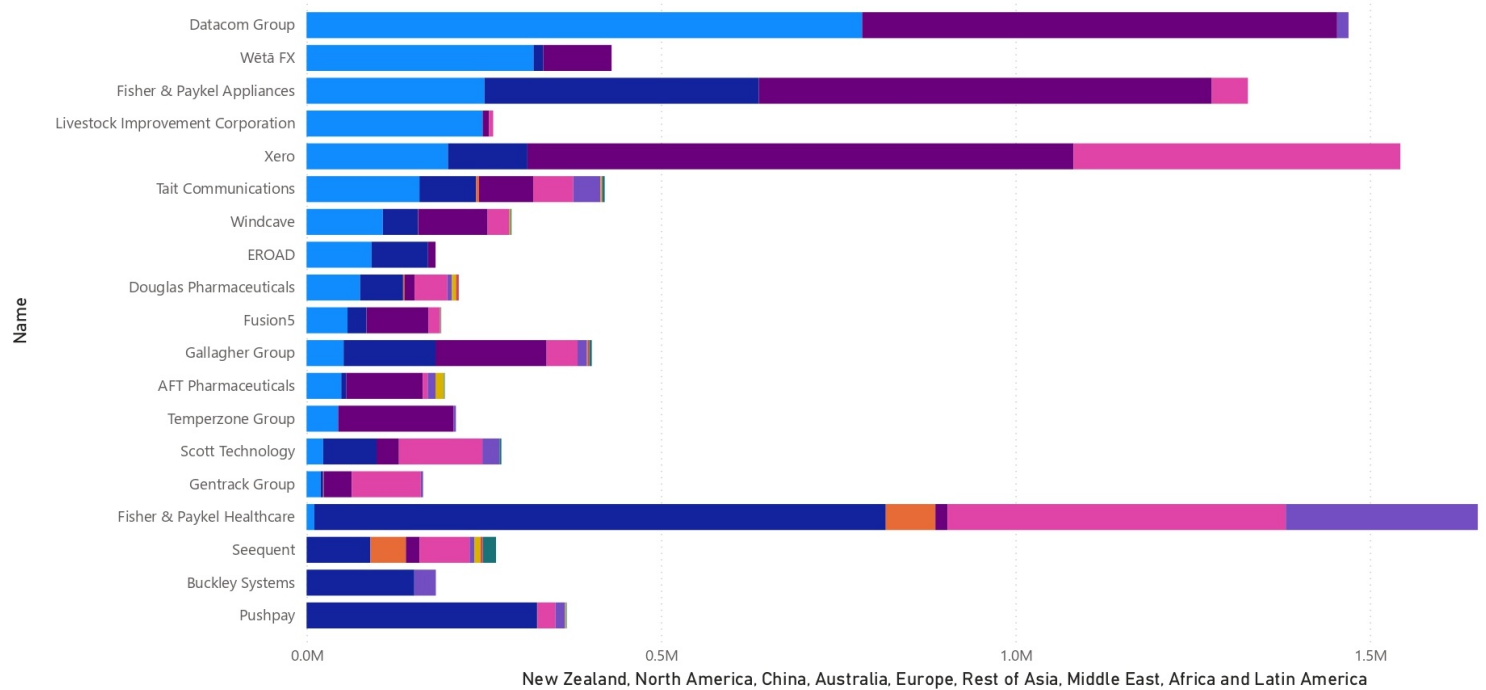
TIN200 Companies by Ownership Type



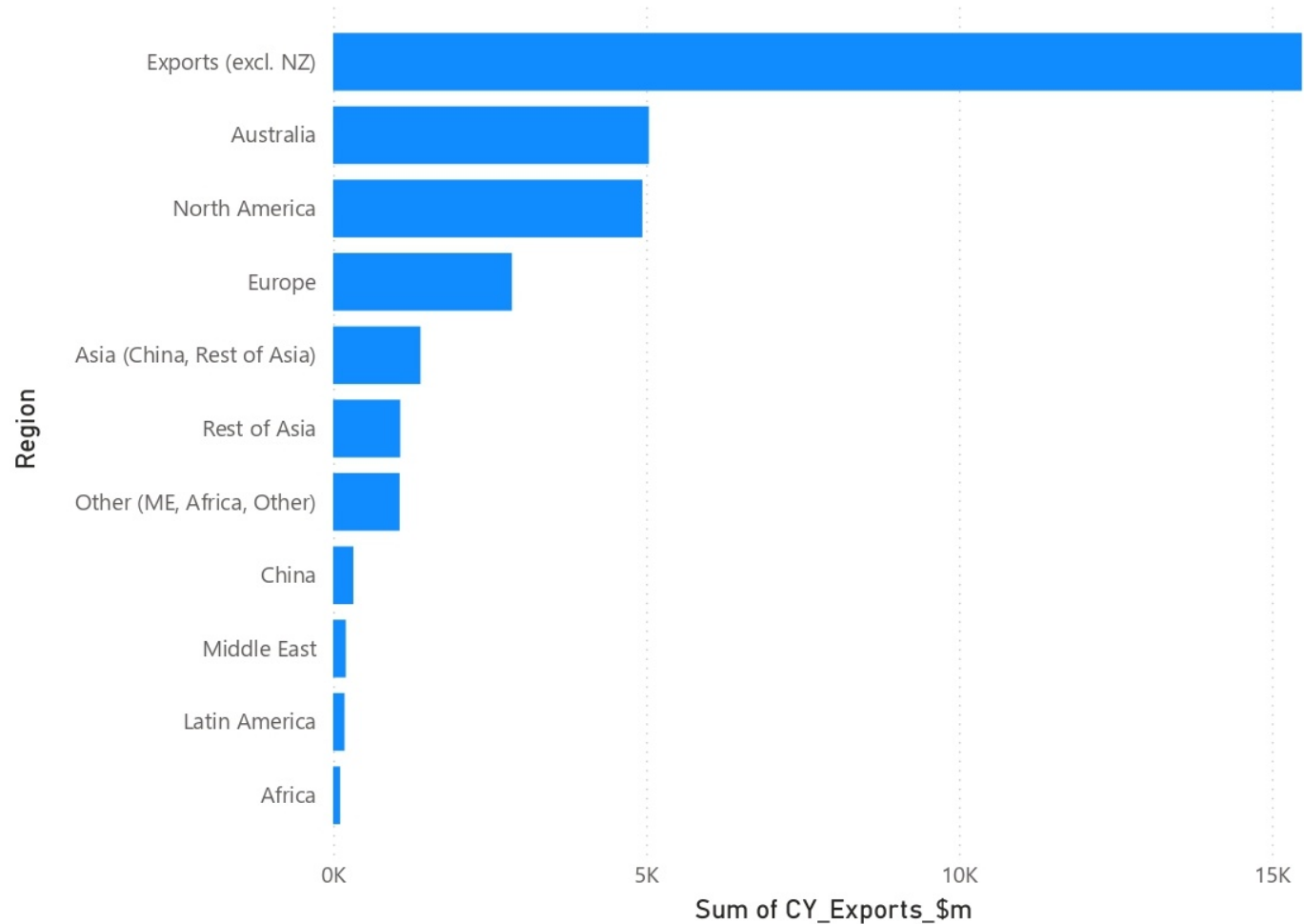
Interpretation. The paired charts separate representation by company count from economic contribution by revenue. Foreign-owned companies account for 50 of 200 companies, while public companies contribute the largest revenue share in the displayed model.

Figure F6. Top-company revenue split by market, 2025**Top 20 Companies: Global Revenue Split 2025 (NZD \$000)**

● New Zealand ● North America ● China ● Australia ● Europe ● Rest of Asia ● Middle East ● Africa ● Latin America



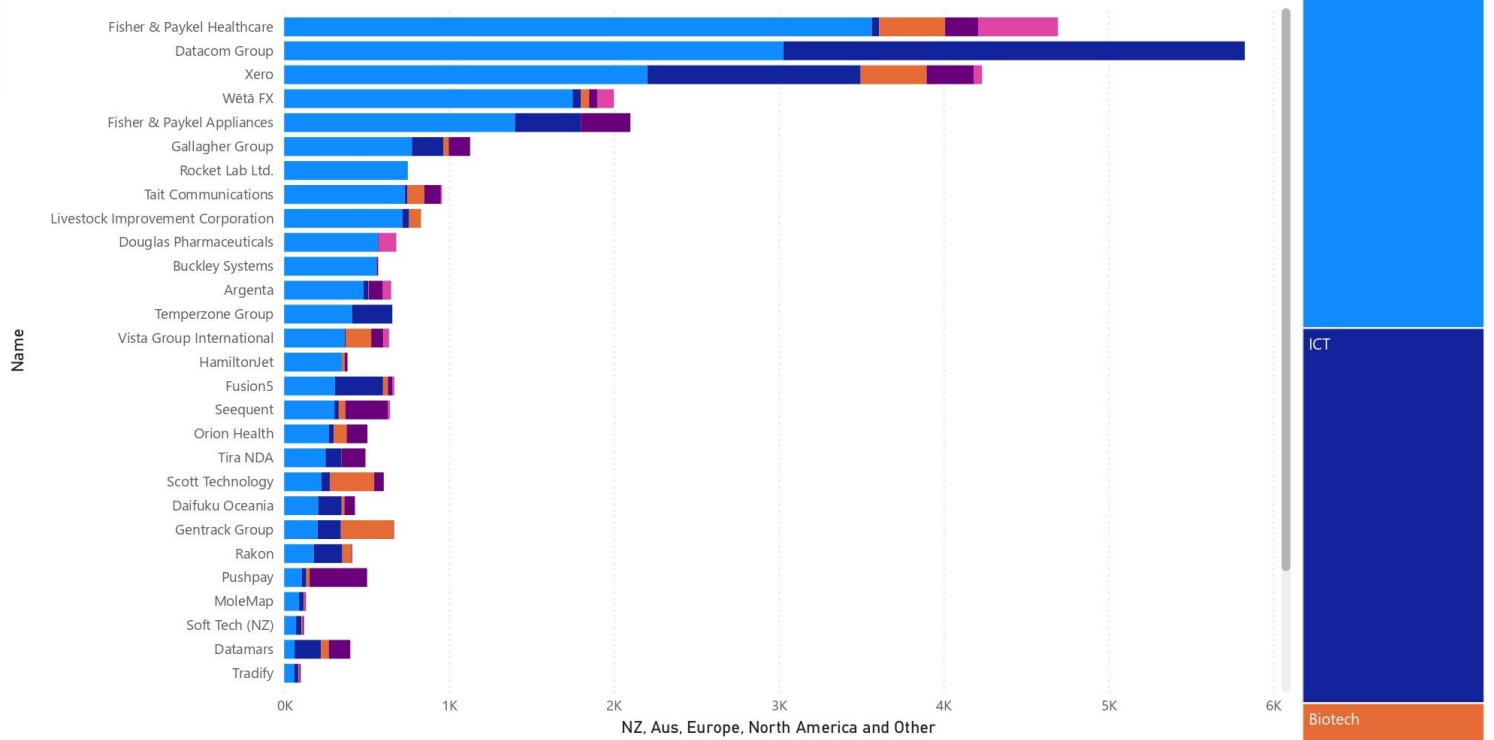
Interpretation. The stacked bars show that international market exposure varies substantially across companies. The visual supports comparing domestic revenue with major offshore markets without treating all exporters as having the same geographic profile.

Figure F7. Technology exports by destination region, 2025**NZ Tech Exports by Destination Region 2025 (\$m)**

Interpretation. Australia and North America are the largest named destination regions in the displayed data, followed by Europe. The first bar is the total export measure and should not be added to the regional components.

Figure F8. Employee distribution by company, geography, and sector**TIN200 Total Employees by Countries & Sector (2025)**

● NZ ● Aus ● Europe ● North America ● Other



Interpretation. The workforce view shows the geographic distribution of employees for selected companies and provides sector context. It highlights that company scale and offshore employment patterns vary considerably across the TIN200.

Source for Figures F1-F8: Power BI report developed from the cleaned TIN project workbooks. Values and labels reflect the report model at the time the screenshots were exported.

Appendix G. Economic Context: The TIN200 and the New Zealand Economy

This appendix places the company data validated during the engagement in its broader economic context. It uses figures reported in TIN's official 2025 release and transparent calculations shown in Table G1. The indicators provide context for the importance of accurate company records; they do not replace company-level revenue verification or TIN's ranking methodology.

Appendix G.1 Sector Scale and Growth

TIN reported NZ\$20.0 billion of combined TIN200 revenue in 2025, up 9.9% year on year, with exports rising 12.4% to NZ\$15.31 billion (Technology Investment Network, 2025). Compared with NZ\$8.95 billion in 2015, the 2025 revenue figure represents total growth of approximately 123.5% and a ten-year compound annual growth rate of approximately 8.4%. The calculation is shown in Table G1. The scale and sustained growth of the population reinforce the need for consistent company identity, eligibility, and ownership records.

Appendix G.2 Revenue Concentration and Scale

TIN's official 2025 release states that 40 companies generated more than NZ\$100 million of revenue. This equals 20% of the 200-company population by count, but the release does not provide enough information to calculate the share of total revenue generated by each revenue band. No additional band-level concentration percentages are therefore presented. The verified result still indicates the presence of a substantial group of large firms while preserving the distinction between company count and revenue concentration.

Appendix G.3 Sector and Regional Patterns

High-tech manufacturing was the largest primary sector in 2025 at NZ\$9.80 billion, followed by ICT at NZ\$8.93 billion and biotech at NZ\$1.25 billion (Technology Investment Network, 2025). Relative to the reported NZ\$20.0 billion total, these values represent approximately 49.0%, 44.7% and 6.3%, respectively. The figures are rounded and sum to NZ\$19.98 billion, which is consistent with the rounded total reported by TIN.

TIN reported NZ\$10.5 billion of revenue for Auckland-based companies and NZ\$5.2 billion for Wellington-based companies in 2025 (Technology Investment Network, 2025). These values are approximately 52.5% and 26.0% of the reported total. Together, the two regions account for about 78.5% of TIN200 revenue. No additional regional-growth or headquarters-distribution inference is made because the retained evidence did not include the necessary schedules.

Appendix G.4 Derived Indicators and Interpretation

Two useful indicators can be calculated directly from TIN's official 2025 figures. Exports of NZ\$15.31 billion, divided by NZ\$20.0 billion in revenue, yield an export-intensity ratio of approximately 76.6%. Revenue of NZ\$20.0 billion divided by 61,369 employees yields approximately NZ\$325,900 per employee. Revenue per employee is a scale indicator, not a formal labor-productivity measure, because it uses revenue rather than value added. Comparisons with national productivity and R&D intensity are not possible because the retained report lacked comparable definitions and calculation schedules.

Table G1. Calculation schedule for derived economic indicators

Indicator	Inputs	Formula	Result	Evidence source
Ten-year revenue growth	2015 revenue = NZ\$8.95b; 2025 revenue = NZ\$20.0b	$(20.0 / 8.95) - 1$	123.5%	Technology Investment Network (2025)
Ten-year revenue CAGR	2015 revenue = NZ\$8.95b; 2025 revenue = NZ\$20.0b; n = 10	$(20.0 / 8.95)^{(1/10)} - 1$	8.4%	Same inputs as above
Export intensity	Exports = NZ\$15.31b; revenue = NZ\$20.0b	$15.31 / 20.0$	76.6%	Technology Investment Network (2025)
Revenue per employee	Revenue = NZ\$20.0b; employees = 61,369	$\text{NZ\$20.0b} / 61,369$	NZ\$325,900	Technology Investment Network (2025)
Companies above NZ\$100m	40 companies; population = 200	$40 / 200$	20.0%	Technology Investment Network (2025)
Primary-sector shares	High-tech = 9.80b; ICT = 8.93b; biotech = 1.25b; total = 20.0b	Sector revenue / 20.0	49.0%; 44.7%; 6.3%	Technology Investment Network (2025)
Selected regional shares	Auckland = 10.5b; Wellington = 5.2b; total = 20.0b	Regional revenue / 20.0	52.5%; 26.0%	Technology Investment Network (2025)

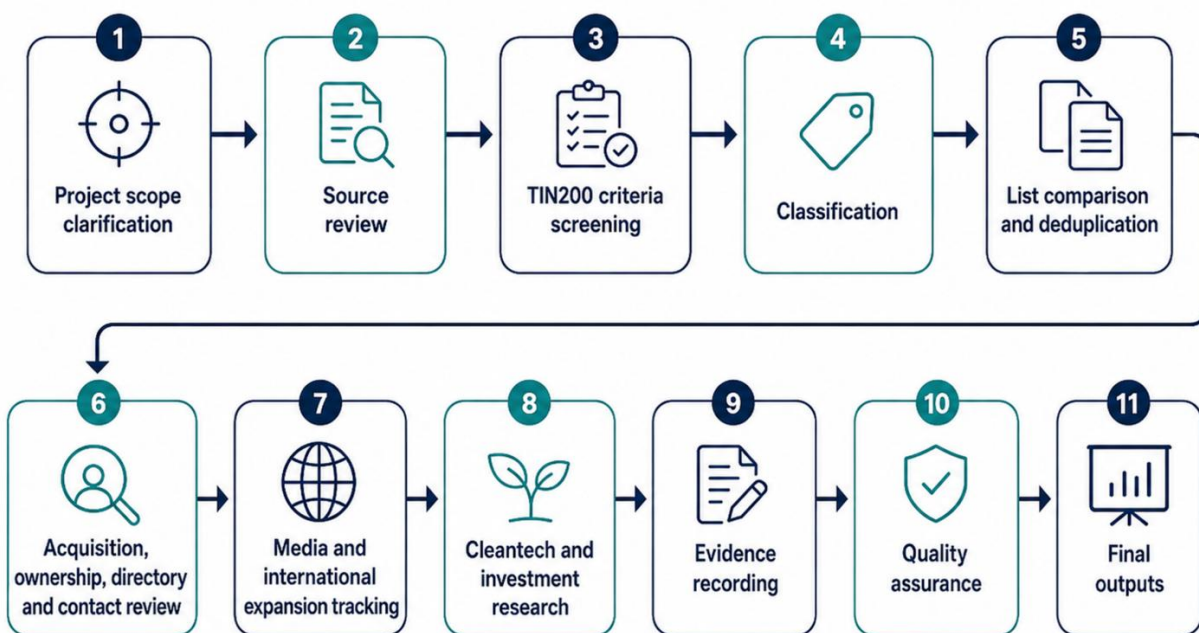
Note. Values are rounded to match TIN's published figures. Percentages may not sum exactly because the published sector values and total are rounded. The calculations are descriptive and should not be interpreted as independent forecasts or formal productivity estimates.

Appendix H. Supplementary Pipeline and Market Research Evidence

This appendix preserves the strongest additional content from the broader team report: aggregate candidate screening, contact-readiness, media-monitoring, and cleantech mapping outputs. The populations are deliberately kept separate from the audited denominators in the main findings. They support discovery, prioritization, and future research planning, but they do not prove final TIN200 eligibility or revenue rank.

Appendix H.1 Supplementary workflow sequence

Figure H1. Broader project research and validation sequence



Interpretation. The figure describes the complete set of team activities. It complements Figure 1, which presents the recommended recurring governance control rather than the historical project sequence.

Appendix H.2 Population map and evidence boundaries

Table H1. Distinct workstream populations

Workstream	Population	Recorded result	Use in this report
Core candidate validation	40 candidate records	31 Keep; 6 Review; 3 Remove	Audited workbook used in the principal findings
Broader screening workstream	127 working records	59 shortlist; 55 manual review; 13 excluded	Supplementary aggregate; not pooled with the 40-record workbook
Existing-list comparison population	207 comparison records	Reference population reported in the broader screening workstream	Supplementary context, distinct from the 143-company reference used for the 34 sponsor entries
Media monitoring	63 refined records reviewed	49 retained in-period; 14 outside 2025-2026	43 source-backed and 6 supported by a generic source page, as classified by the team
Cleantech mapping	92 companies or records	Segment and stage profiles retained	Pipeline and ecosystem monitoring only; not evidence of TIN200 eligibility

Evidence boundary. The 127-record screening population, the 207-record comparison population, and the 40-record candidate-validation population must not be combined or treated as a single funnel. They reflect different inputs and decisions.

Appendix H.3 Broader candidate-screening profile

Table H2. Broader screening disposition

Category	Count	Interpretation
Shortlist / proceed	59	Retained for outreach, further review, or working-list consideration
Manual review	55	Public evidence, qualification, contact quality, or overlap requires checking
Excluded	13	Removed for duplication, prior listing, or a recorded exclusion reason
Total working records	127	Distinct broader screening population

Table H3. Sector profile of the 59-record shortlist

Sector	Companies
SaaS	8
AgTech	5
HealthTech	5
FinTech	5
Climate Tech	4
Software	4
Cybersecurity	3
CleanTech	2
Advanced Manufacturing	2
Life Sciences	2
Other sectors	19
Total	59

Interpretation. The shortlist is diverse, and no single sector dominates. The 19-record 'Other sectors' category should be decomposed before management reporting because it currently hides nearly one-third of the shortlist.

Appendix H.4 Contact-readiness profile

Table H4. Contact availability by screening category

Category	Companies	Email or email-style contact	Phone contact	Named person
Final shortlist	59	37 (62.7%)	23 (39.0%)	7 (11.9%)
Needs review	55	31 (56.4%)	19 (34.5%)	3 (5.5%)
Excluded	13	7 (53.8%)	5 (38.5%)	0 (0.0%)

Interpretation. Generic contact details were substantially more common than named role-relevant people. Contact readiness should therefore remain a separate operational status and should not be interpreted as evidence that a company satisfies the TIN criteria.

Appendix H.5 Media-monitoring outputs

Table H5. Media-monitoring verification summary

Measure	Result
Original refined records reviewed	63
Records retained in the 2025-2026 period	49
Records excluded as outside the period	14
Records classified as source-backed	43
Records supported by a generic source page	6

Table H6. Retained records by development category

Development category	Records
Funding announcements	4
Acquisitions	11
International expansion	6
New product launches	13
Significant leadership changes	5
Strategic partnerships	10
Total	49

Interpretation. Product launches, acquisitions, and strategic partnerships formed the largest categories. These events can trigger a directory, ownership, description, or contact review, but an announcement is not proof of revenue, export share, or final TIN200 eligibility.

Appendix H.6 Cleantech discovery universe

Table H7. Cleantech records by segment

Segment	Records
Waste and recycling	16
Climate software	16
Sustainable agriculture	14
Renewable energy	12
Carbon management	12
EV and transport	9
Water technology	4
Solar	3
Hydrogen	3
Battery materials	1
Battery storage	1
Other/unclassified	1
Total	92

Table H8. Cleantech records by development stage

Development stage	Records
Early	40
Growth	19
Early growth	10
Established	7
Early / R&D	4
Scale-up	3
Growth/infrastructure	2
Public/global public company	4
Other	3
Total	92

Interpretation. The mapping is useful as an emerging-company watchlist: 40 of the 92 records were classified as early stage, and a further 14 as early growth or early/R&D. Monitoring should focus on evidence of commercialization, offshore revenue, New Zealand presence, and independent financial reporting rather than investment visibility alone.

Appendix H.7 Recommended use and limitations

The supplementary workbooks should be retained as research assets and linked to the central candidate register through stable company identifiers. Screening disposition, contact readiness, current developments, and ecosystem segment should be stored as separate fields. Media events should carry a date, event type, evidence link, and verification status. Cleantech records should carry stage, segment, New Zealand connection, export evidence, and next-review date.

The aggregate tables were reproduced from the team report and should remain in the final submission only alongside the underlying confidential workbooks. The report does not reproduce personal email addresses, unsupported company-level funding claims, subjective labels such as 'best' or 'most promising', or investment amounts that were not accompanied by row-level evidence. Those items can remain in internal research files pending verification.

Used this way, the supplementary research expands TIN's discovery capability without weakening the decision boundary established in the main report: public information and market signals can justify monitoring or outreach, while final TIN200 inclusion remains dependent on TIN's five criteria, company evidence, and revenue ranking.